



Fundamentals 3 – Chemical Properties and Adhesion

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Chemical properties are integral to understanding material stability

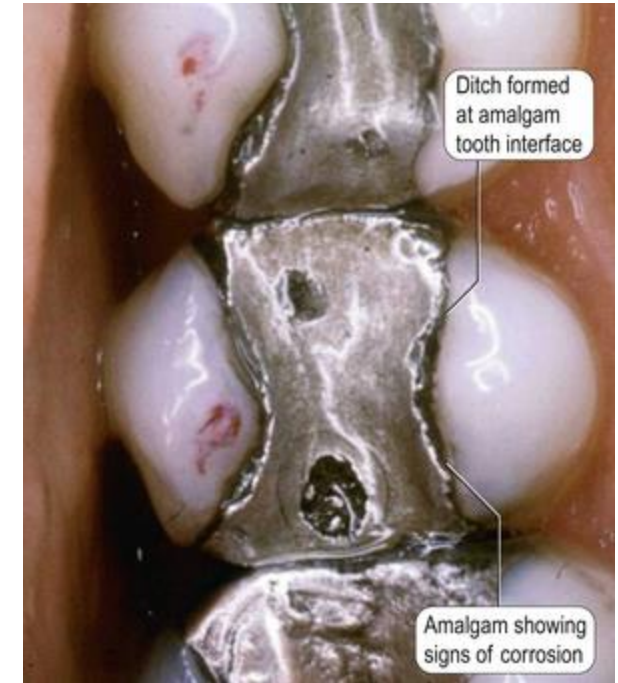
- Polymers – suffer from absorption and loss of soluble components
- Metals – prone to tarnish and corrosion
- Ceramics – chemical dissolution

Degradation of polymers

- Solvent molecules, usually water, can be absorbed and force polymer chains apart --> reduces intermolecular bond strength --> material swells, softens, and weakens
 - Example:
 - In resin composites: gradual discoloration from porosity formation, hydrolysis of resin-filler interface, and dissolution of filler
- Polymeric materials absorb water (weakens bonds) **and** lose water-soluble components that support the materials' properties

Tarnish and corrosion of metals

- Tarnish – surface-level discoloration due to hard/soft deposits, can be removed by polishing
- Corrosion - chemical reaction leading to degradation of metal, leading to fracture
 - Corrosion products may also lead to undesired reactions with biological environment



Wet Corrosion

- Metallic atom (anode) loses an e^- to O_2 dissolved in water (cathodic process)
- Can sometimes be *localized* instead of a uniform occurrence

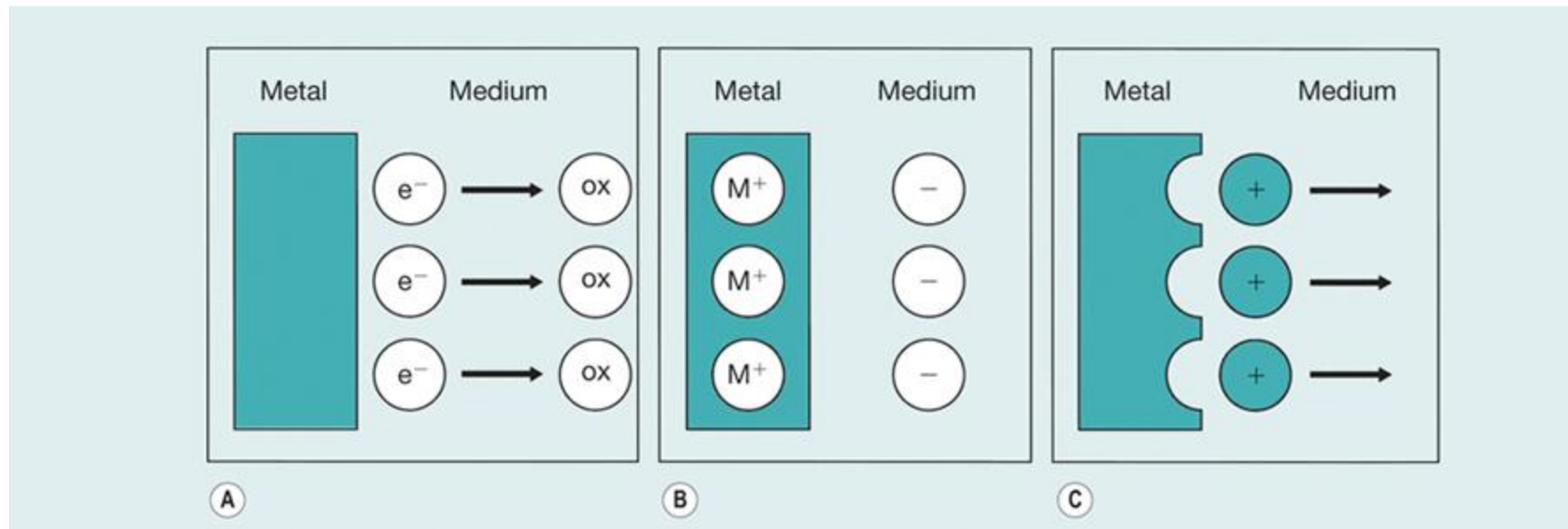


TABLE 1.8.1

The Galvanic Series in Seawater

↑	Platinum Gold Titanium Silver
Increasingly inert	Stainless steel Copper Nickel
Increasingly active	Tin
↓	Lead Cast iron Aluminium Zinc

Galvanic corrosion

- Occurs when 2 metals, or metallic phases, contact each other, causing in increased corrosion of one of the metals
 - Creates a galvanic cell

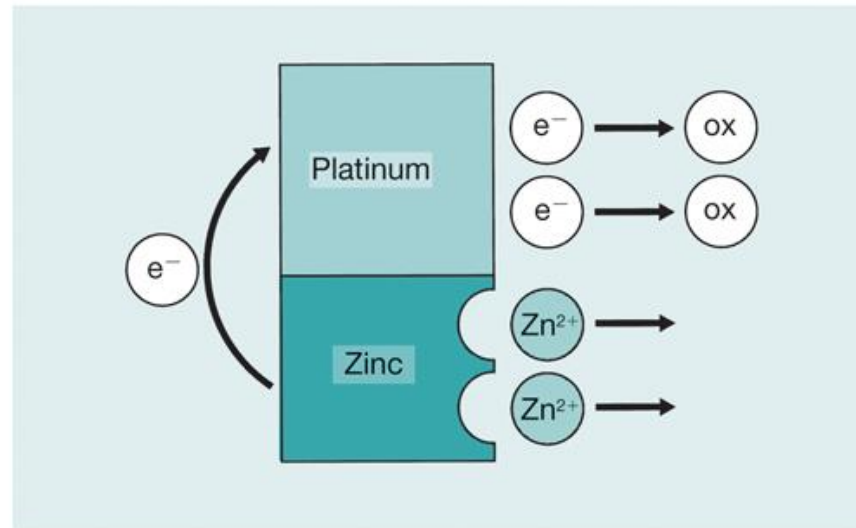


FIGURE 1.8.5 Galvanic corrosion of zinc in contact with platinum. The noble metal (platinum) is more easily oxidized than the base metal (zinc), such that the anodic process takes place at the zinc surface and zinc ions are released into the aqueous environment.

Crevice corrosion

- Oxygen is absent in the crevice, so e^- must travel to surface
- Metal / corrosive products are released in the crack, further restricting oxygen and propagating the process
 - Eventually leads to material failure

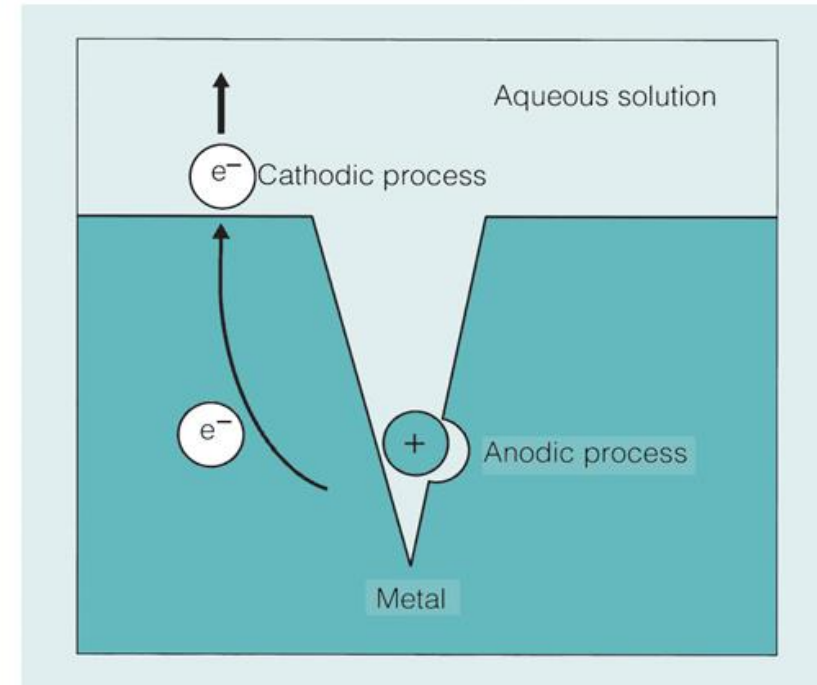


FIGURE 1.8.6 In crevice corrosion the oxidation takes place at the surface (cathodic process) and metal ions are released from within the crevice (anodic process).

Degradation of ceramics

- Usually chemical - components may dissolve in water or acids
- Physical stresses can expedite chemical corrosion, leading to ceramic failure

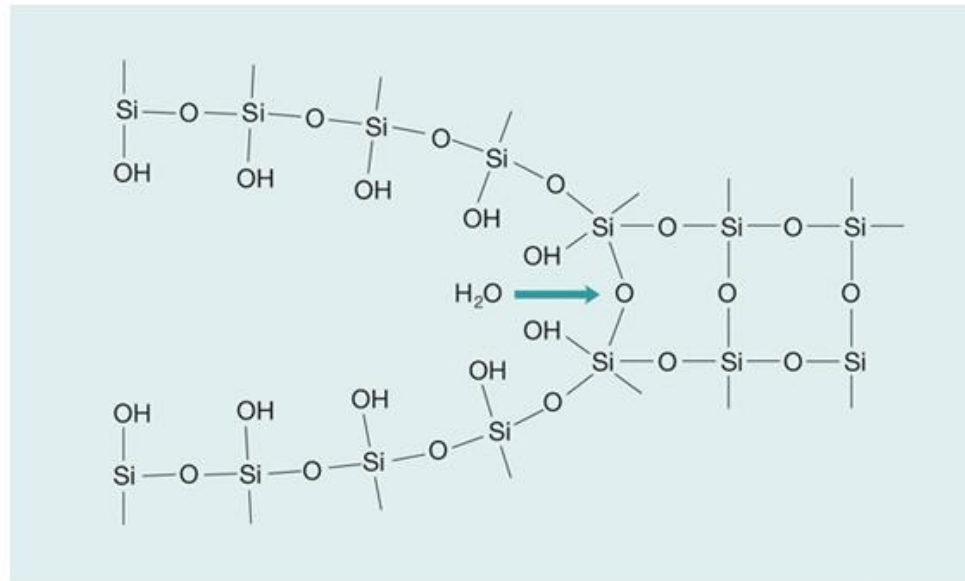
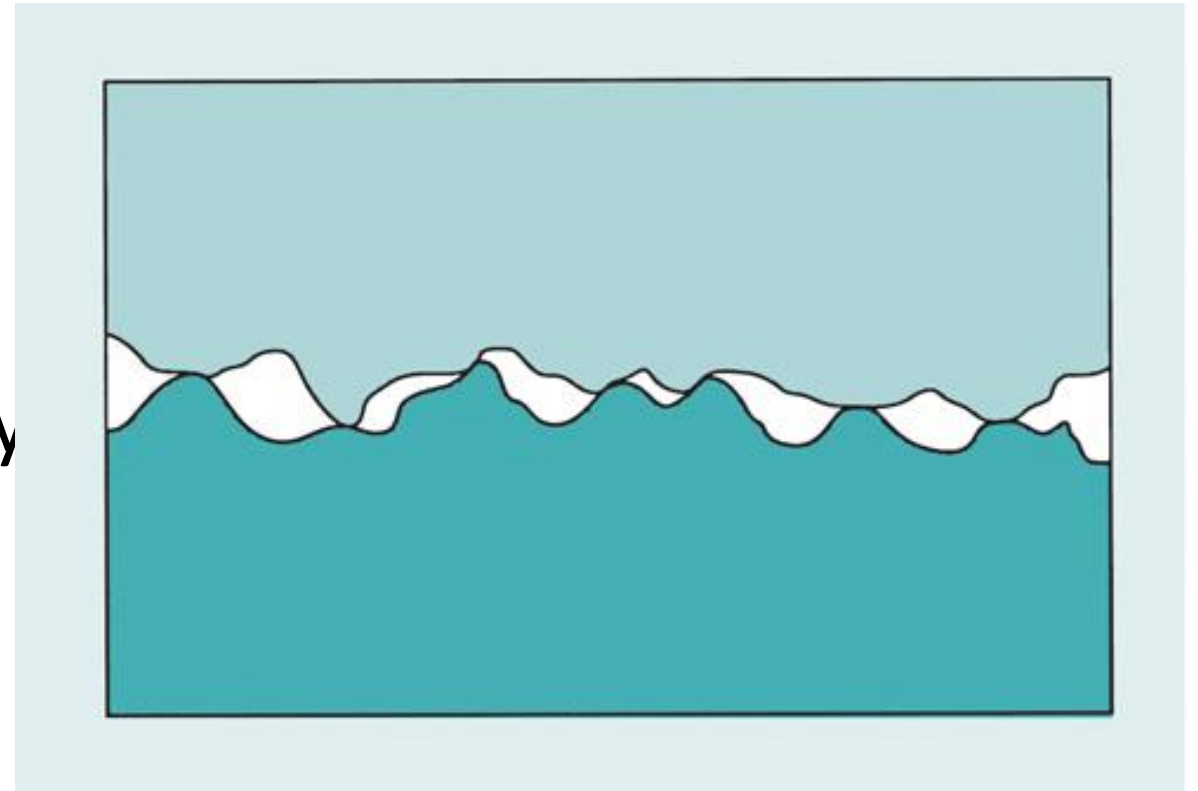


FIGURE 1.8.7 Crack growth in a ceramic due to local hydration at the tip of the crack.

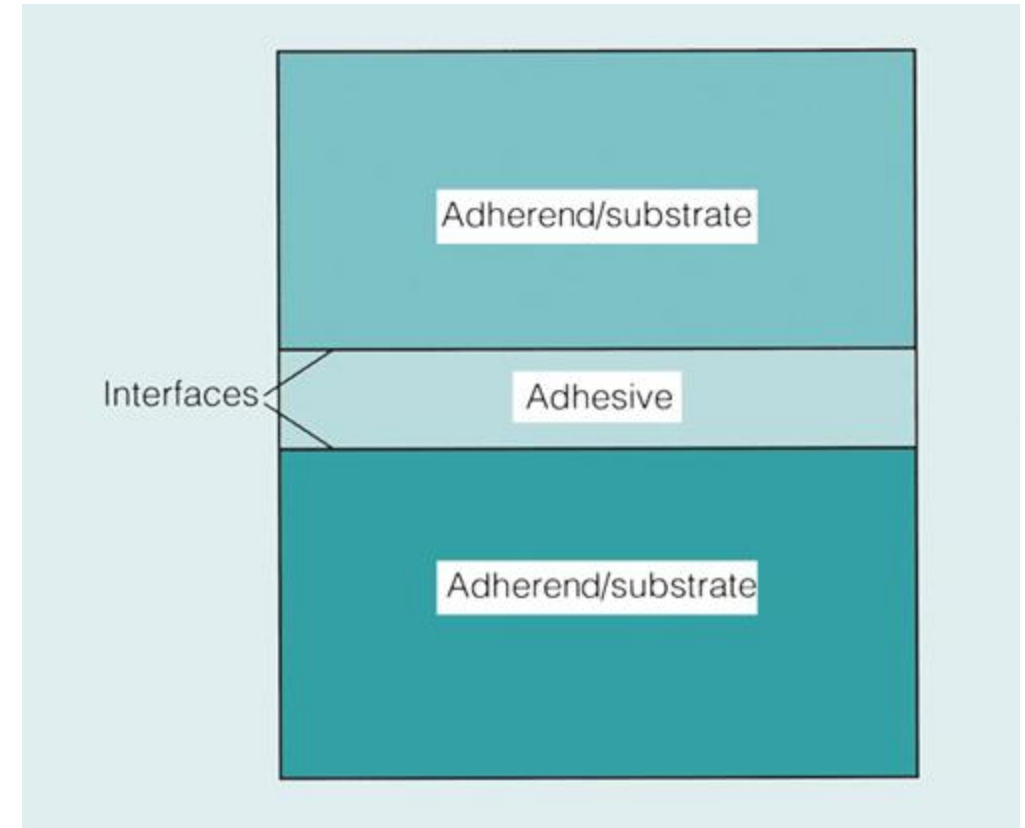
Principles of adhesion

- Force that binds 2 dissimilar materials together when in contact
- Solid-solid adhesion is virtually impossible since adhesion requires intimate contact of molecules, and surfaces are macroscopically rough



Solid-solid adhesion with liquid adhesive

- Liquids create intimate contact over a solid's surface, strengthening molecular attraction via van Der Waals forces and making adhesion possible
 - Example: water droplets on any surface
- Thus, to create adhesion of two solids, you need a fluid or semi-fluid **adhesive** that adheres to both solids
- Need dry and clean solid surfaces to maximize probability of adhesion



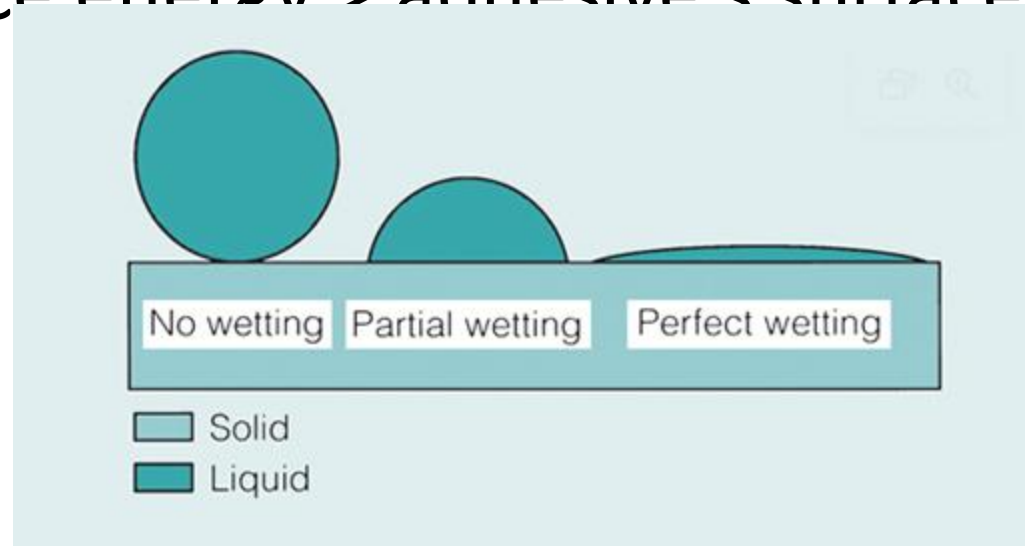


Adhesive properties to maximize adhesion

- Wettability of substrate by adhesive
- Viscosity of adhesive
- Morphology or surface roughness of substrate

Wettability

- The ability of an adhesive to create full, intimate contact with a surface without air voids
- Surface energy, the strength of inward pull of surface-molecules, is the main factor
- Substrate surface energy > adhesive's surface energy allows better wetting



Viscosity

- Viscosity inhibits wetting a surface and penetration into narrow cracks/crevices

Surface roughness

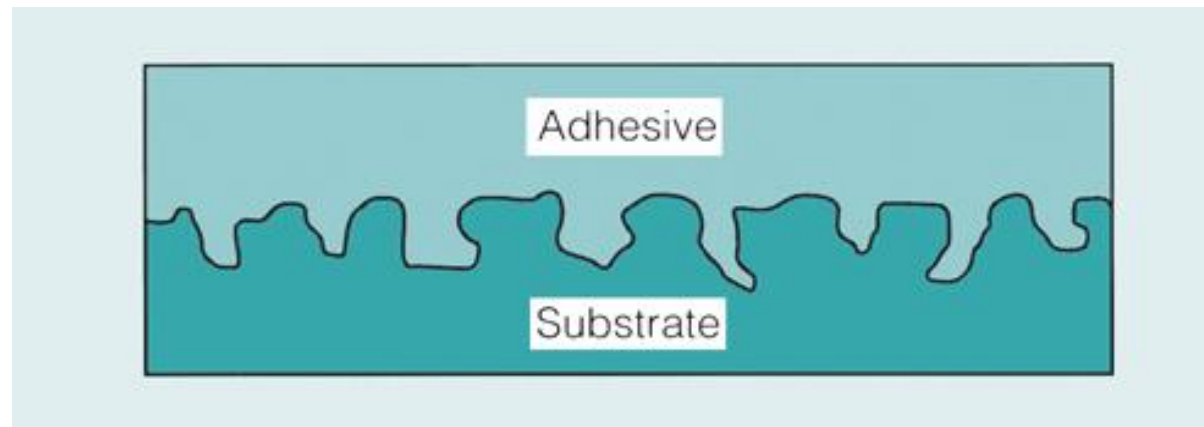
- Rough surfaces provide more surface area for bonding, but also more areas for air/contaminant entrapment
 - Highly viscous adhesives will have trouble filling in cracks or crevices
- If the rough surface irregularities have certain morphology, such as undercuts, micromechanical interlocking can be achieved

The ideal adhesion scenario

- Adhesive's surface energy/tension is lower than substrate's
- Adhesive has low viscosity
- Substrate has a rough surface, with undercut-shaped irregularities

Mechanical adhesion

- Micromechanical interlocking into undercuts on substrate surface
- Penetration of adhesive into undercuts dependent on the pressure used in application and on the properties of the adhesive
- Not dependent on molecular attractive forces



Chemical adhesion

- Covalent bonding between adhesive's atoms and substrate's surface
- Sometimes, the bond is strong enough to entangle the molecular chains of the adhesive and the substrate

Interplay of adhesive (bond) strength and cohesive strength

- Cohesive forces are those that keep similar molecules within a substance in close proximity
 - Reminder: adhesive forces keep 2 dissimilar molecules together
- External stresses (i.e. tensile forces, compressive forces, etc.) are applied to dental restorations constantly
- What will generally happen when adhesive forces are greater than the cohesive forces within a restoration?

Contributors to bonding failure (debonding)

- Improper wetting of surface due to contaminants
- Adhesive layer is too thick
 - Adhesives do not have as strong mechanical properties (tensile and shear strength) as substrates
 - Thicker adhesive layers shrink more, reducing intimate surface coverage and adhesive strength
 - Then external forces on the tooth/substrate will lead to breakdown of the adhesion
 - Thus, dentists want a thin layer of bond (minimal film thickness)

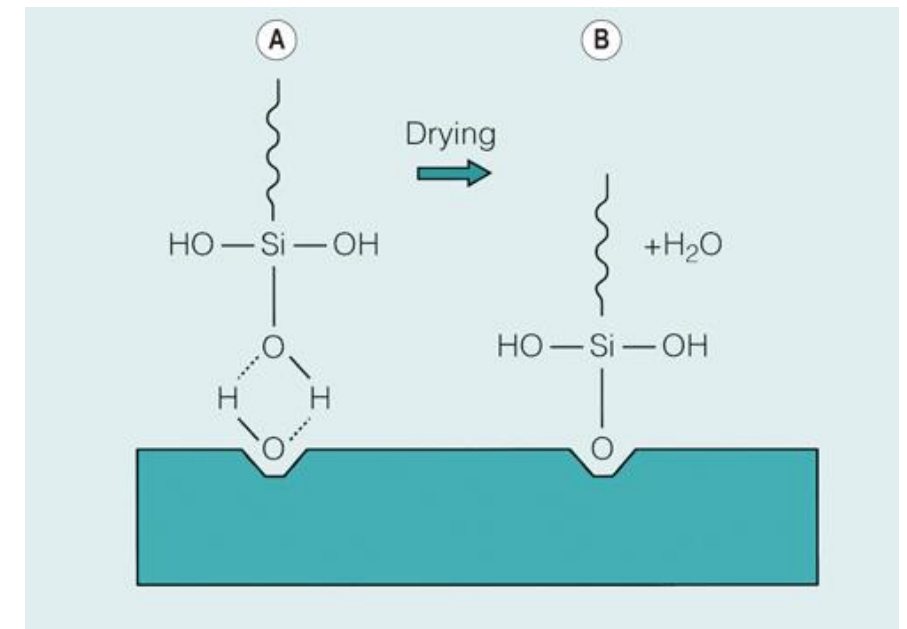


Adhesion promoters

- Needed when two substances do not have a strong affinity for each other
 - Example: the resin and glass filler in resin composite materials
- Coupling agent - intermediary substance that can bond to both substances
- Primer - modifies one of the substances to allow bonding

Coupling agents

- Glass and polymeric resin do not naturally adhere together
 - Water naturally adheres to glass surfaces, preventing glass-resin bonding
 - In composite restoration material, glass-resin bonding is necessary so that the glass can be load-bearing with the resin
- Coupling agents (silanes) displace the water on glass surfaces, allowing H-bonding between glass surface oxides and resin polymers



Primers

- Changes surface of a substrate to be more bondable
- Examples: phosphoric acid etch, dentin conditioners